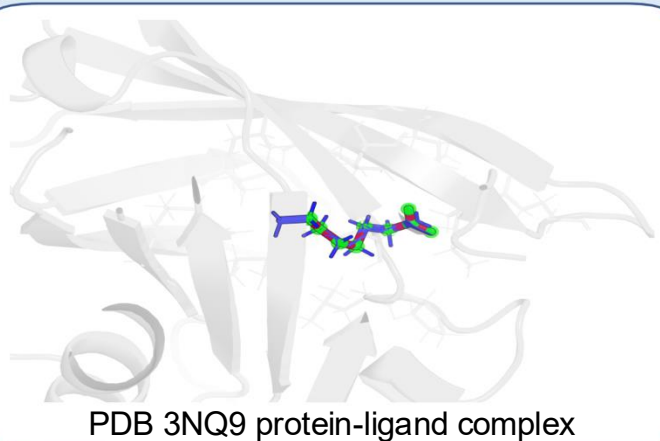
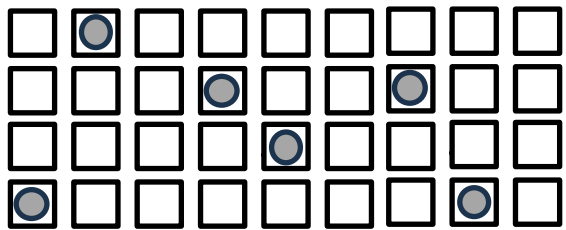


STEP 1

Discretization



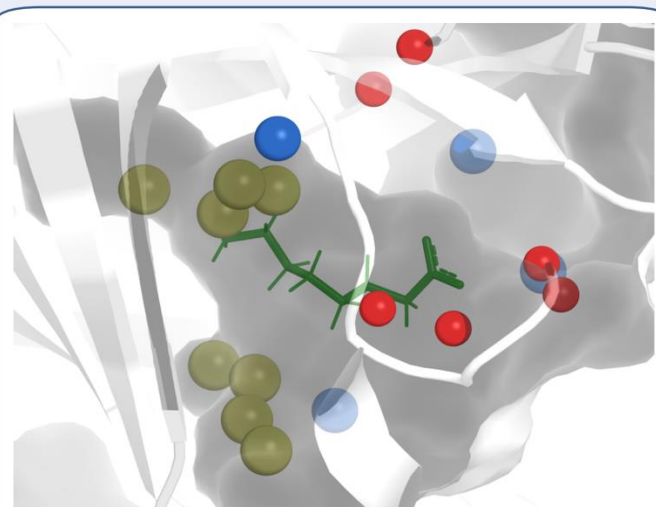
AutoSite probes on 3D grid



Pocket grid → candidate
cell coordinates

STEP 2

Feature Atom Extraction



- H-acceptor (OA)
- H-donor (HD)
- hydrophobic (C)

AutoDock Receptor {C,HD,OA} map
→ top-N lowest-energy grid cells per type

3NQ9 yields 13 features / 4JSZ yields 12

STEP 3

Ligand Atom Mapping (FAM)

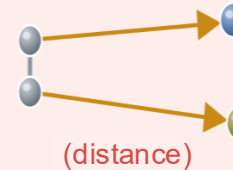
a_i, a_k (ligand atom) g_j, g_l (feature atom)

valid FAM matching



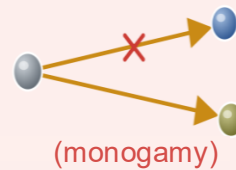
valid if $|d_{ik} - D_{jl}| \leq \delta_{dist}$ and each atom $\leftrightarrow$ 1 feature

✗ stretched



(distance)

✗ multi-mapped



(monogamy)

$$\begin{cases} x_{ij} = 1 & \text{if } a_i \rightarrow g_j \\ x_{ij} = 0 & \text{otherwise} \end{cases} \begin{cases} \text{monogamy constraints} \\ \text{distance constraints} \end{cases}$$

Penalize via $E_{mono} + E_{dist}$

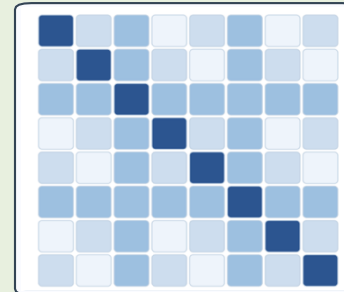
STEP 4

QUBO Formulation

Eq. (10) FAM-QUBO energy

$$E(x) = E_{score}(x) + K_{mono}E_{mono}(x) + K_{dist}E_{dist}(x)$$

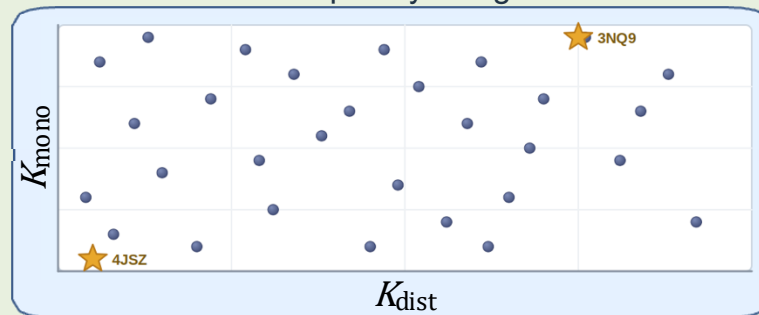
QUBO matrix Q



K_{dist} penalty
distance consistency

K_{mono} penalty
monogamy (1-to-1)

BO penalty tuning



★ marks BO-discovered optimum per target